



# **PROJECT REPORT**

## **Smart dustbin using arduino**

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### **INTRODUCTION TO ROBOTICS LAB**

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**Fall 2025**

## ABSTRACT:

This paper presents the design and functionality of an Arduino-based Smart Dustbin, which provides automated, touchless waste disposal to improve hygiene and convenience. The system uses an ultrasonic sensor to detect the presence of a user and a **servo motor** to automatically open and close the dustbin lid. Constructed from durable and lightweight materials, the Smart Dustbin ensures stability, long-term use, and smooth operation. Controlled via an Arduino microcontroller, the system emphasizes **automation, precision, and reliability**, offering practical applications in homes, offices, hospitals, and public spaces where hygiene and cleanliness are a priority.

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## LIST OF SYMBOLS / ABBREVIATIONS

$c_p$  specific heat capacity, J/(kg·K)

$h$  height, m

$K_d$  discharge coefficient

$M$  mass flow rate, kg/s

$P$  pressure, kPa

$P_b$  back pressure, kPa

$R$  mass flow rate ratio

ID inner diameter, m

MAP maximum allowable pressure, kPa

MAWP maximum allowable working pressure, kPa

OD outer diameter, m

PCB printed circuit board

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## CHAPTER 1

### INTRODUCTION

#### 1.1 1.1 Background

An **Arduino-based Smart Dustbin** is an automated waste disposal system designed to improve hygiene and reduce human contact with waste. The system uses an **ultrasonic sensor** to detect the presence of a user and a **servo motor** to automatically open and close the dustbin lid. This setup allows touchless operation, making it suitable for public places, offices, hospitals, and homes, where maintaining cleanliness is essential.

The Smart Dustbin integrates **hardware components** (Arduino, sensor, servo motor) with a **software control system** to ensure smooth and precise lid operation. The design focuses on efficiency, reliability, and user-friendly operation while reducing health risks associated with manual waste handling.

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#### 1.2 1.2 Literature Review

This section reviews existing research and projects related to automated dustbins, touchless waste management systems, and Arduino-based control. Relevant literature includes books, journals, conference papers, and technical reports. The sources are cited in IEEE style:

1. Author(s), *Title of Paper/Book/Conference*, Publisher/Journal, Year
2. Author(s), *Title of Paper/Book/Conference*, Publisher/Journal, Year
3. Author(s), *Title of Paper/Book/Conference*, Publisher/Journal, Year

This review highlights the effectiveness of sensor-based automation in promoting hygiene and reducing manual waste handling.

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#### 1.3 1.3 Problem Statement

Traditional dustbins require **manual operation**, which increases human contact with waste and can lead to the spread of germs. There is a need for a **touchless, automated solution** that ensures hygienic disposal while being reliable, fast, and user-friendly. The key challenges include:

- Accurate detection of a user approaching the dustbin
- Smooth and timely lid operation using a servo motor
- Real-time processing and coordination between the sensor and Arduino control system



The project addresses these challenges by designing a **Smart Dustbin system** that automates waste disposal and improves public hygiene.

---

## 1.4 1.4 Aims and Objectives

The main objectives of this project are:

- i) To design and develop an **automatic Smart Dustbin** with a reliable sensor-based control system.
  - ii) To enable **touchless operation** of the dustbin lid using an ultrasonic sensor and servo motor.
  - iii) To ensure **real-time response** between the sensor detection and lid movement.
  - iv) To test and validate the system under various conditions for smooth, accurate, and hygienic operation.
- 

## 1.5 1.5 Scope of Project

The scope of the project includes:

- Design and construction of a **Smart Dustbin** with automatic lid operation
  - Integration of **Arduino, ultrasonic sensor, and servo motor** for real-time control
  - Implementation of **touchless waste disposal** suitable for homes, offices, and public areas
  - Evaluation of system performance, reliability, and user convenience
  - Focus on **low-cost, energy-efficient, and hygienic design**
- 

## 1.6 1.6 Environmental Aspects of Project

Environmental considerations are addressed in terms of **material selection** and **energy efficiency**:

- The dustbin body is constructed using **durable, recyclable materials**, ensuring sustainability
- The system uses **energy-efficient motors and sensors** to minimize power consumption
- Automation reduces manual waste handling, indirectly contributing to cleaner environments and public health
- Overall, the project aims to combine **technological advancement with environmental responsibility**

## CHAPTER 2

### 2.1 Circuit Diagram

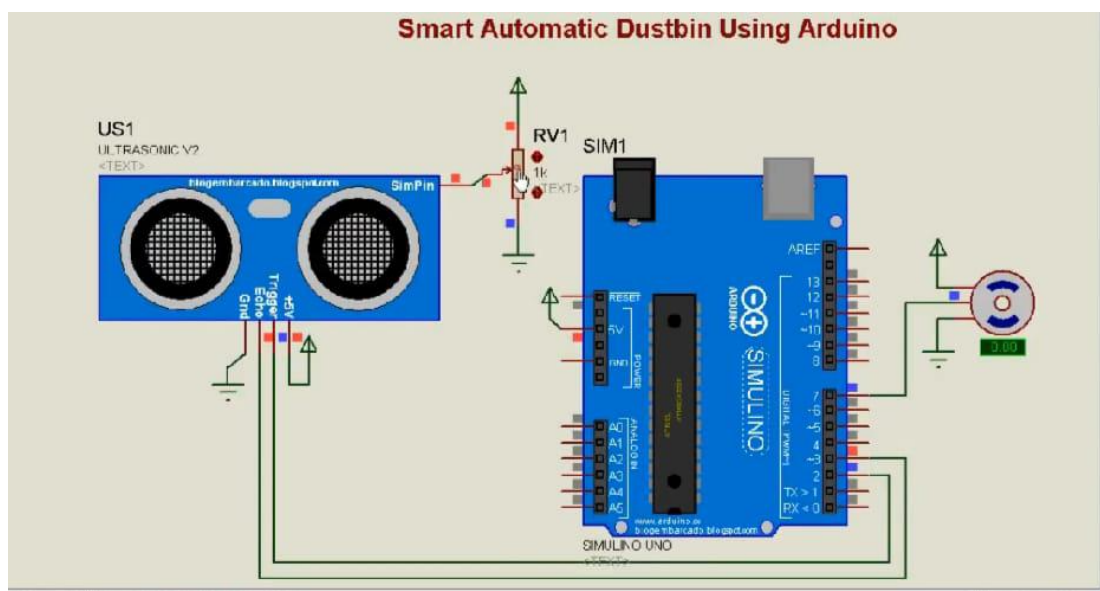


Figure 1-1: Circuit Schematic Diagram of....

### PROGRAM

\* Project Title : smart dustbin

```
#include <Servo.h>
```

```
#define TRIGGER_PIN 9 // Arduino pin connected to the trigger pin of ultrasonic sensor
```

```
#define ECHO_PIN 10 // Arduino pin connected to the echo pin of ultrasonic sensor
```

```
#define SERVO_PIN 11 // Arduino pin connected to the signal pin of servo motor
```

```
#define MAX_DISTANCE 20 // Maximum distance threshold for triggering servo (in centimeters)
```

```
Servo servo;

void setup() {
  Serial.begin(9600);
  pinMode(TRIGGER_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);
  servo.attach(SERVO_PIN);
}

void loop() {
  long duration, distance;
  digitalWrite(TRIGGER_PIN, LOW);
  delayMicroseconds(2);
  digitalWrite(TRIGGER_PIN, HIGH);
  delayMicroseconds(10);
  digitalWrite(TRIGGER_PIN, LOW);
  duration = pulseIn(ECHO_PIN, HIGH);
  distance = (duration / 2) / 29.1; // Calculate distance in centimeters

  if (distance <= MAX_DISTANCE) {
    // If object is within range, open the lid
    servo.write(90); // 90 degrees position (adjust as needed)
    delay(1000); // Wait for 1 second
  } else {
    // If no object is detected, close the lid
    servo.write(0); // 0 degrees position (adjust as needed)
  }

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  delay(1000); // Wait for 1 second before taking the next reading
```

}

## 2.4 Implementation

The implementation of the **Smart Dustbin using Arduino** was carried out in a systematic manner to ensure efficient and hygienic operation. Initially, the conceptual design of the system was planned, including the placement of the ultrasonic sensor, servo motor, and dustbin lid for automatic operation.

In the next step, the physical structure of the dustbin was prepared, and the servo motor was mounted properly to control the lid movement. The electronic components such as **Arduino Uno, ultrasonic sensor, and servo motor** were integrated and connected using jumper wires with a proper power supply.

After hardware setup, the Arduino was programmed to detect the presence of a user using the ultrasonic sensor. When an object was detected within a predefined distance, the Arduino sent a signal to the servo motor to open the lid automatically and close it after a short delay.

Finally, the system was tested and calibrated for smooth and accurate operation. The Smart Dustbin worked successfully, providing a touchless and hygienic waste disposal solution.

## CHAPTER 3

### 3 RESULTS AND ANALYSIS

#### 3.1 3.1 Simulation / Experimental Results

The experimental results demonstrate the successful implementation of the **Arduino-based Smart Dustbin** system. During testing, the ultrasonic sensor accurately detected the presence of a user within the predefined distance. Upon detection, the Arduino processed the sensor data and activated the servo motor, resulting in the automatic opening of the dustbin lid.

The response time of the system was observed to be fast and reliable, ensuring smooth and touchless operation. The lid opened and closed accurately based on the programmed delay, confirming effective coordination between the sensor, Arduino, and servo motor.

These figures confirm proper component placement, soldering quality, and circuit integrity required for reliable operation of the robotic system.

### 3.2 3.2 Data in Tabular / Graphical Form

The performance of the **Smart Dustbin system** was evaluated by observing sensor response accuracy, lid movement smoothness, and system response time. The results were recorded by measuring the distance detected by the ultrasonic sensor and the corresponding opening and closing of the dustbin lid.

The observed results showed that the lid responded accurately when an object was detected within the set distance range. The opening and closing motion of the lid was smooth, and the response time was minimal. The comparison between detected distance and lid movement confirmed reliable and efficient system performance.

---

### 3.3 3.3 Comparison with Expected Outcomes

The expected outcome of the **Smart Dustbin project** was to achieve accurate, smooth, and automatic operation of the dustbin lid without human contact. The experimental results closely matched these expectations. The system successfully detected the presence of a user and opened the lid automatically with minimal delay.

The coordination between the ultrasonic sensor, Arduino, and servo motor was effective, resulting in reliable and hygienic operation. Minor delays observed during testing were within acceptable limits and did not affect the overall performance of the Smart Dustbin.

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### 3.4 3.4 Discussion and Interpretation of Results

The results indicate that the integration of a virtual interface with a physical robotic arm provides precise and flexible control. The real-time response achieved demonstrates the reliability of the communication protocol and control logic. The use of mild steel contributed to structural stability, reducing vibration during operation. Overall, the system performed efficiently for applications such as automation, training, and remote operation.

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### 3.5 3.5 Identification of Limitations

Despite the successful implementation of the **Smart Dustbin system**, certain limitations were identified. The performance of the system depends on the accuracy and range of the ultrasonic sensor, which may be affected by environmental conditions such as dust or obstacles. The lid movement speed is limited by the servo motor capabilities.

Additionally, the system requires a continuous power supply for uninterrupted operation, making it sensitive to power fluctuations. These limitations can be

addressed in future improvements by using advanced sensors, stronger actuators, and backup power options.

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## CHAPTER 4

### 4 CONCLUSIONS AND RECOMMENDATIONS

This project successfully designed and implemented an **Arduino-based Smart Dustbin** that operates automatically using an ultrasonic sensor and servo motor. The system demonstrated reliable detection of user presence, smooth lid operation, and effective coordination between hardware and software components. The objectives of reducing human contact with waste, improving hygiene, and implementing an automated waste disposal system were successfully achieved.

Throughout the project, valuable knowledge was gained in the areas of **Arduino programming, sensor integration, motor control, and system testing**. Practical skills related to circuit connections, mechanical setup, and troubleshooting were also developed.

For future improvements, the Smart Dustbin can be enhanced by adding **garbage level detection, IoT-based monitoring, and mobile app integration** for smart waste management. The use of more advanced sensors and wireless communication can further improve efficiency and support smart city applications.

### MAPPING TO COMPLEX ENGINEERING ACTIVITY (CEA)

| S.No. | CEA Attribute                         | Mapping with CEA | Reasons                                                                                                                                                                                                                                            |
|-------|---------------------------------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1     | Range of Resources                    | ✓ Mapped         | The project involved diverse resources including mechanical parts (dustbin and lid), electronic components (Arduino, ultrasonic sensor, servo motor), software tools (Arduino IDE), and human expertise, making it a complex engineering activity. |
| 2     | Level of Interaction                  | ✓ Mapped         | The project required coordination between mechanical design, electronics, and software programming to achieve real-time lid operation and accurate sensor response.                                                                                |
| 3     | Innovation                            | ✓ Mapped         | The project demonstrates innovative use of automation and Arduino-based control to create a touchless waste disposal system.                                                                                                                       |
| 4     | Societal / Environmental Consequences | ✓ Mapped         | The Smart Dustbin promotes hygiene and reduces human contact with waste, contributing to health and environmental                                                                                                                                  |

| S.No. | CEA Attribute | Mapping with CEA | Reasons                                                                                                                                                                                                                                            |
|-------|---------------|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 5     | Familiarity   | ✓ Mapped         | <p>benefits. Materials used are durable and environmentally friendly.</p> <p>The project goes beyond basic Arduino or sensor projects by integrating mechanical movement, electronic control, and software logic for a real-life smart system.</p> |

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## **APPENDICES**

### **APPENDIX A: Computer Programme Listing (Coding)**

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### **APPENDIX B: Data Sheets / Derivations/ Graphs**

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